

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83) -	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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APN 038050009000 | ZONING AP | ACRES 0.481 ac

BEST GNSS WINDOW Sun, Jul 12 · 2:00 PM – 6:00 PM | PDOP 0.97 · 35 sv | NEAREST CORS
P261 · 14.1 mi ✓

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Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 12, 2026

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

SITE & SIGNAL REPORT

COORDINATES

38.3459, -122.3047

WGS84 / NAD83

ELEVATION

66 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.0° E

Compass reads 13.0° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.6

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

20.6 TECU

-5.2 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.6). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Sun, Jul 12	0.97	2:00 PM – 6:00 PM	35	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates. For site-specific weather, terrain mask angle, and nearest CORS station, order an On-Demand Site Report.



WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 12, 2026) · Conditions may change – verify before fieldwork



GPS CONSTELLATION HEALTH

▲ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.9–0.9 across survey window
PDOP remains ≤ 0.9 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Sunday, Jul 12

WEEKEND

Overcast

10 mph

GO

High 90°F Low 56°F Wind max 10 mph Gusts 12 mph Precip 0.00" Sunrise 5:56 AM Sunset 8:33 PM

Waning Crescent 6%

GNSS: Avg PDOP 0.97 · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.93, 35 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	0.87	36 sv	IDEAL
7:00 AM	0.91	34 sv	IDEAL
8:00 AM	1.06	29 sv	GOOD
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.00	29 sv	GOOD
11:00 AM	1.06	27 sv	GOOD
12:00 PM	0.96	29 sv	IDEAL
1:00 PM	1.00	28 sv	GOOD
2:00 PM	0.90	34 sv	IDEAL
3:00 PM	0.98	33 sv	IDEAL
4:00 PM	0.98	34 sv	IDEAL
5:00 PM	0.86	38 sv	IDEAL
6:00 PM	1.01	32 sv	GOOD
7:00 PM	1.04	30 sv	GOOD



GO - GOOD CONDITIONS

Partly cloudy | Wind: 9.5 mph (gusts 11.6 mph) | Precip: 0" | Temp: 56.3°–89.7°F



DRONE WIND ANALYSIS



GO Peak: 9.5/11.6 mph SSE

* Sunrise: 5:56 AM Sunset: 8:33 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≠turbulence	DIR
5 AM	3/5 mph		SSE
6 AM	4/7 mph		ESE
7 AM	2/3 mph		ESE
8 AM	3/3 mph		W
9 AM	3/4 mph		SW
10 AM	5/5 mph		SSW
11 AM	6/7 mph		SSE
12 PM	5/6 mph		S
1 PM	4/5 mph		SE
2 PM	6/7 mph		S
3 PM	7/8 mph		S
4 PM	8/10 mph		S
5 PM	10/11 mph		S
6 PM	9/11 mph		SSW
7 PM	9/12 mph		S
8 PM	6/11 mph		SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#) ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

JOB: N/A | SITE: N/A | LOCATION: 38.3459, -122.3047 (Napa County)

REPORT GENERATED: Sunday, July 12, 2026

VERDICT

GO — PDOP geometry is excellent (0.93 at peak window), weather is stable with no precipitation, and ionospheric activity is moderate with acceptable delays for static work. Survey conditions are favorable.

BEST SURVEY WINDOW

2:00 PM – 6:00 PM local time Sunday, July 12, 2026

PDOP: 0.93 | Satellite count: 35 (GPS+GLONASS+Galileo+BeiDou)

Rationale: Peak geometry window with lowest dilution of precision and maximum constellation strength. Overcast conditions present no operational threat; wind (10 mph max) is negligible.

BEST 4-HOUR STATIC GNSS WINDOW

2:00 PM – 6:00 PM (full window recommended)

PDOP remains at or near 0.93 throughout. This 4-hour occupation is optimal for high-accuracy static positioning and baseline resolution. Moderate ionospheric activity (Kp 4.7, TEC 20.6 TECU) is manageable at this window length. Double-frequency receivers will mitigate residual ionospheric delay effectively.

DRONE OPERATIONS ASSESSMENT

ACCEPTABLE — Overcast conditions and 10 mph max wind are within safe operating parameters for most UAS platforms. Visual line of sight operations pose no weather risk. No airspace conflicts noted. Operations permitted throughout daylight window if required; 2:00 PM – 6:00 PM recommended to align with peak GNSS geometry for any ground-truthing or tie-in work.

FIELD SAFETY WARNINGS

MINOR — Moderate geomagnetic activity (Kp 4.7) may cause transient multipath or cycle slip on single-frequency receivers in open-sky environments. Mitigation: use dual-frequency receivers and extend occupation windows by 10–15% if single-frequency equipment is deployed. No other hazards identified. Air quality is good (AQI 45). No active wildfires detected within range. Temperature range 56–90°F is operationally acceptable.

NARRATIVE SUMMARY

Survey conditions on Sunday, July 12, 2026, at Napa County location 38.3459, -122.3047 are favorable for GNSS static work. PDOP geometry is excellent (0.93 at peak), with 35-satellite constellation available 2:00 PM – 6:00 PM. Weather is stable (overcast, no precipitation, 10 mph wind). Moderate ionospheric activity (Kp 4.7, F10.7 138 sfu) is within acceptable range for static surveys; use dual-frequency receivers and normal occupation protocols. GPS PRN 11 is unhealthy and will be auto-excluded by receivers; this does not degrade PDOP materially given constellation strength. Site conditions include dwarf scrub cover with low multipath risk and recommended 13° elevation mask. Wildfire proximity all-clear; air quality good. Proceed with survey as planned during the 2:00 PM – 6:00 PM window for optimal results.

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk
1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor
1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P261	HUNTERHILLCN2004	14.1 mi
 NGS ↗	P198	PETALUMAIRC2004	17.5 mi
 NGS ↗	OHLN	OHLONE PARK	23.5 mi
 NGS ↗	P196	MEACHUMLFLCN2006	24.0 mi
 NGS ↗	CASR	SANTA ROSA CA	24.9 mi
 NGS ↗	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet ↗	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

Copy All

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	697.01 ft (-9 ft)	706 ft	36K af	68%
Folsom Lake	66.5 mi	448.64 ft (-17 ft)	466 ft	792K af	81%
Lake Oroville	93.4 mi	862.11 ft (-38 ft)	900 ft	2,871K af	81%
New Melones Lake	100.1 mi	1,020.25 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	55%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	697 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL
004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa Valley Cashman Formation is a Quaternary alluvial and lacustrine deposit that fills the structural basin of Napa Valley, formed by sediment erosion from surrounding highlands during multiple climate cycles. This formation typically contains interbedded sand, gravel, silt, and clay lenses that create multiple water-bearing zones at varying depths. At this site, water is readily available within 10–90 feet of the surface, but yields and quality will depend on which specific sand or gravel layer a well penetrates.

FORMATION

Name	Napa Valley Cashman Formation
Age / Rock	Quaternary alluvial and lacustrine sand, gravel, silt, and clay
Extent	250–350 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	51–295 ft (5–30 floors)
Typical Depth	80–150 ft (8–15 floors)
Recharge Source	Precipitation on valley floor and tributary alluvial fans; lateral groundwater from bedrock uplands
Transit Time	5–20 years to 100 ft depth

WATER QUALITY

pH	est. 7.0–7.6 (est. USGS/DWR)
Hardness	est. 150–350 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 250–500 mg/L (est. USGS/DWR)

Notes	No sample data available. Typical Napa Valley alluvial water is moderately hard and suitable for most uses; local artesian conditions possible in deeper zones.
TREND	
Trend	Water levels show significant long-term decline (1977–2022): shallow measurements average ~24 ft BGS (1977), deepest well 87.71 ft BGS (2014). Single 2022 measurement (21.86 ft) suggests possible recovery but insufficient data for definitive trend.
Source	Site well records; pattern consistent with USGS/DWR regional Napa Valley groundwater monitoring
Seasonal Swing	est. 10–25 ft (based on Napa Valley regional patterns; site-specific data limited)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ **LOW SUBSIDENCE RISK — No known critically overdrafted basin**
Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU	
The North Coast Coastal Ranges aquifer system (N100CACSTL) in this region is characterized by well-drained soils and consolidated bedrock geology typical of California's coastal mountainous terrain, which provides natural resistance to subsidence. The area is not designated as a critically overdrafted groundwater basin under SGMA, indicating sustainable groundwater conditions with minimal large-scale pumping stress. For surveyors working in this area, historical benchmarks and elevations are expected to remain stable without requiring subsidence corrections to survey data.	
SUBSIDENCE ASSESSMENT	
Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin
BENCHMARK IMPLICATIONS	
Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate
SURVEY ACTION	
Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)		WIND / SNOW / ICE / FROST (ASCE 7-22)	
Site Class	D (default)	Basic Wind Speed	92 mph
Risk Category	II	Ground Snow Load	5.8 psf
		Winter Wind (W2)	0.35
		Ice Thickness	None
		Frost Depth	12 in minimum
		Est. from IECC Zone 3C — AFI data unavailable	
		Verify with local building official before construction	
		Source: ASCE 7-22 Hazard Tool · gis.asce.org	

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

🕒 AI CLIMATE SUMMARY — RESIDENTIAL DESIGN	
This marine climate features mild temperatures year-round with moderate coastal winds at 92 mph, so your building envelope should prioritize air sealing and impact-resistant glazing, particularly on the windward (typically western) facades. Concentrate operable windows and glazing on sheltered eastern and northern exposures to capture passive heating in winter while minimizing solar gain and wind exposure on the ocean-facing side. The key passive design strategy for this zone is using thermal mass (concrete, tile, or masonry) on south-facing interior surfaces to moderate the modest but consistent temperature swings, combined with natural ventilation through cross-winds during the frequent mild periods to reduce cooling loads.	
🕒 AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU	
This coastal marine site presents well-drained gravelly loam soils suitable for conventional spread footing design. Seismic parameters are currently unavailable and should be confirmed with the local building official; wind exposure is moderate (92 mph) and anticipated to require standard design per ASCE 7-22. Frost depth of approximately 12 inches and marine climate conditions may warrant attention to drainage and corrosion protection in foundation design.	
FOUNDATION	
Type	Conventional spread footing anticipated, subject to geotechnical confirmation
Basis	Well-drained Coombs gravelly loam soils suggest adequate bearing capacity for spread footings; consult licensed engineer to confirm bearing pressure and settlement estimates
STRUCTURAL IMPLICATIONS	
Seismic	Seismic design category and response parameters currently unavailable—consult local building official to confirm SDC assignment and any anticipated seismic detailing requirements
Wind Exposure	Exposure C (open terrain with scattered obstructions) suggested per ASCE 7-22; verify site topography and surrounding land use
Wind	92 mph 3-second gust wind speed anticipated—standard wind load design likely adequate for Risk Category II occupancy; no hurricane exposure anticipated for this inland location
Snow	5.8 psf ground snow load anticipated—standard roof design and drainage provisions likely adequate; consult engineer for flat or low-slope conditions

INVESTIGATIONS & INSPECTIONS

Geotech Required	Yes—geotechnical investigation recommended to confirm Site Class D assignment, bearing capacity, settlement estimates, and frost depth verification
Liquefaction Screen	Screening recommended—confirm shallow groundwater depth and soil saturation conditions with licensed geotechnical engineer, particularly given Site Class D and well-drained soil profile
Special Inspections	Dependent on final SDC confirmation; consult engineer once seismic parameters are established
Code Edition	ASCE 7-22 / IBC 2021 (planning reference only)
Site Class Note	Site Class D assumed for planning purposes—geotechnical investigation strongly recommended to confirm soil profile, density, and saturation conditions

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes	SERIES: Coombs
DRAINAGE CLASS: Well drained	HYDRO GROUP: C
SOIL ORDER: Alfisols	SUBORDER: Xeralfs

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is an Alfisol (Xeralf suborder) formed in alluvial and colluvial materials derived from mixed rock sources in the California Coast Ranges and interior valleys. This well-drained soil features a gravelly loam surface layer overlying an argillic (clay-enriched) substratum typical of Mediterranean climates with summer drought. Site developers and surveyors should expect moderate gravel content (15–35%), good trafficability during dry seasons, and seasonal moisture dynamics that influence fill placement and foundation design.

Alluvial and colluvial mixed-origin parent materials on gentle upland slopes and fan remnants; annual grassland and oak savanna vegetation.

DRAINAGE

Class	Well drained
Seasonal Saturation	None within 60 inches
Flooding Risk	Minimal on 2–5% slopes
Ponding Risk	None

ENGINEERING

Bearing Capacity	2,500–3,500 psf (shallow foundations)
Rating	Fair
Shrink-Swell	Moderate — argillic clay substratum in Xeralfs
Frost Action	Low
Cut Suitability	Good (gravel-rich gravelly loam layer)
Fill Suitability	Fair (clay substratum — moisture control and compaction required)

SEPTIC

Suitability	Conditional
Notes	Perc test required — clay-enriched argillic horizon may restrict drainage; minimum 3–4 feet separation to clay layer recommended

HAZARDS

Excavation	Moderate — 15–35% gravel content may increase equipment wear; moderate rippability
Expansive Clay	Moderate — argillic horizon typically 18–35% clay; monitor in dry season shrinkage
Erosion	Low on 2–5% slopes; gully and rill initiation risk if surface disturbed during wet season
Other	None

Data sources: California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdnldataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America ·
Real-time data subject to revision
Water levels current as of report run: Jul 12, 2026, 9:59 PM UTC
* Live pool elevation not reported by this reservoir — full pool design elevation shown

OIL & GAS WELLS California — within 10mi radius

8 total
0 active
7 plugged
 By status:
 7 Plugged
 1 Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well, a plugged Mobil Oil Corporation dry hole operated by Mobil Oil Corporation located 7.2 miles away, is well beyond any actionable distance threshold. No active producers, injection wells, or other wells within relevant setback distances require further review.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200
total in search area

10
within 5 mi

108
within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Three active producers operate within 5 miles of the site. Basalt Rock Co (2 mi, clay operations) and Mt George Pumice Deposit (3.1 mi) and Silva Pumice Quarry (5 mi) will generate heavy truck traffic on access roads during operating hours, potential dust emissions, and noise from crushing/blasting operations. No inactive deposits within 1 mile warrant Phase 1 environmental assessment.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6
Endangered

6
Threatened

12
Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	<i>Ambystoma californiense</i>	Endangered

COMMON NAME	SCIENTIFIC NAME	STATUS
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Assessment ****Section 7 Consultation Triggers:**** If the proposed project involves federal permits, funding, or agency approvals, consultation with USFWS is mandatory given the 12 listed species with critical habitat within 25 miles of this site in the North Bay/Sacramento Valley region. ****Most Planning-Relevant Species:**** California tiger salamander, vernal pool tadpole shrimp, and California red-legged frog present the highest regulatory constraints, as they require intact seasonal wetland complexes and upland habitat connectivity—typical constraints on development and grading activities. ****Recommended Next Steps:**** Conduct a Phase 1 biological survey to document existing wetland communities, seasonal inundation patterns, and species presence/absence; then prepare a biological assessment or letter evaluating project-specific impacts and avoidance/mitigation measures in coordination with USFWS. Early agency coordination during planning phases will streamline permitting and identify feasible project alternatives that minimize conflicts with listed species recovery objectives. ***For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.***

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57	5051	2026-07-04
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20
Iturriel's Spear	<i>Triteleia laxa</i>	2026-06-15

COMMON NAME	SCIENTIFIC NAME	LAST OBS
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 38.3459, -122.3047 This site supports a **California oak woodland and chaparral transitional community**, dominated by blue oak and California black oak with understory species like California buckeye, California bay, and diverse herbaceous plants adapted to Mediterranean seasonal drought. The assemblage of moisture-indicator species (Braun's Giant Horsetail, box elder) alongside drought-tolerant natives (Canyon liveforever, Dudleya, California poppy) suggests **variable soil moisture across the site, likely with seasonal seepage or drainage convergence zones** supporting riparian or mesic pockets within drier upland areas. For grading and drainage planning, the presence of horsetail and water-loving species indicates **poorly drained or seasonally saturated microsites that require careful assessment to avoid erosion and habitat disruption**; conversely, the abundance of xeric-adapted species means much of the site drains well and will require erosion control during construction. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		635 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		154 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla aureola		91 obs	
Dasymutilla californica		91 obs	
Dasymutilla sackenii		91 obs	
TOXIC PLANTS			
SPECIES		OBSERVATIONS	
Toxicodendron diversilobum		1100 obs	
Datura wrightii		57 obs	

SPECIES	OBSERVATIONS
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Briefing – Northern California Survey Area Your highest-priority hazards are poison oak (extremely common in this region with 1,439 observations) and Northern Pacific rattlesnakes (635 observations), followed by black widow spiders and various toxic Datura plants. For poison oak, wear long sleeves and pants, learn to identify the three-leaflet pattern, and avoid touching your face until you've washed thoroughly; for rattlesnakes, watch where you step and place your hands, wear boots, and make noise while moving through brush. Be alert for Datura plants (distinctive large flowers and spiky seed pods) along disturbed areas and roadsides—never touch or taste anything unfamiliar. Peak rattlesnake activity runs March through October, and poison oak remains a threat year-round but causes more severe reactions in spring and summer; Datura flowers primarily in late summer through fall. ****Always verify current conditions locally before fieldwork.****

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3
Beam Paths

545ft
Closest Beam

3
Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26
Total Airways

13
Jet Routes

2
Victor Airways

11
Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200 91 66 5 14 6
Total Observations Total Species Birds Mammals Reptiles Amphibians

BIRDS (66 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	9
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melospiza crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Western Screech-Owl	<i>Megascops kennicottii</i>	4
Cooper's Hawk	<i>Astur cooperii</i>	3
Anna's Hummingbird	<i>Calypte anna</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltiriparus minimus</i>	3
+ 51 additional species		

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Black-tailed Jackrabbit	<i>Lepus californicus</i>	4
Coyote	<i>Canis latrans</i>	2
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	9
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	8
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	6
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2
Western Skink	<i>Plestiodon skiltonianus</i>	2
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1

COMMON NAME	Scientific Name	OBS
Pacific Ringneck Snake	Diadophis punctatus amabilis	1
Northwestern Pond Turtle	Actinemys marmorata	1
Common Sharp-tailed Snake	Contia tenuis	1
North American Racer	Coluber constrictor	1
Southern Alligator Lizard	Elgaria multicarinata	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	Pseudacris regilla	5
American Bullfrog	Lithobates catesbeianus	3
Foothill Yellow-legged Frog	Rana boylei	3
Rough-skinned Newt	Taricha granulosa	1
Western Toad	Anaxyrus boreas	1
California Giant Salamander	Dicamptodon ensatus	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in Sonoma County, California supports a diverse wildlife community characteristic of oak woodland and grassland habitat with riparian or wetland components, as evidenced by the presence of wading birds (egrets), aquatic amphibians (chorus frogs, pond turtles), and wetland-associated reptiles. Several species of regulatory concern are present, including the Foothill Yellow-legged Frog and California Giant Salamander (both sensitive to habitat degradation), along with the Northern Pacific Rattlesnake and various native aquatic species that indicate year-round or seasonal water availability supporting intact breeding habitat. The species composition and richness (91 species documented) suggest relatively intact grassland and riparian corridors with minimal to moderate disturbance, though the presence of introduced species (Eastern Fox Squirrel, American Bullfrog, Pond Slider) indicates some human influence and habitat modification. Any site development should consider hydrological connectivity to support amphibian and reptile populations, maintenance of oak and grassland cover, and avoidance of riparian areas that function as critical wildlife corridors and breeding habitat. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 • AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 • IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	→ excellent
7/12	30.3°	98.8%	→ excellent
8/12	33.7°	99.3%	→ excellent
9/12	36.9°	99.8%	→ annual optimal
10/12	39.8°	99.8%	→ excellent
12/12	45°	99%	→ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	△ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	△ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.

Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)

4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 · THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 · TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🔍 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Solar & Energy Planning for Your Home Your location gets about 6.6 hours of useful sunlight per day on average, which is solid for solar panels—think of it like having a consistent part-time job powering your home, stronger in summer (9 hours) and lighter in winter (under 4 hours). Your roof's angle at about 38 degrees is naturally perfect for capturing that sun, but here's what surprises most people: a dark roof absorbs three times more heat than a light-colored one, so choosing a lighter color could cut your summer cooling load by two-thirds without any other changes. The real trouble spot is your west-facing windows, which act like an open oven door in the afternoon—summer sun pouring through them creates four times the heat gain of your south windows, so good shades or awnings there pay for themselves quickly. A mature deciduous tree on your south side is your secret weapon: it'll shade your house for about four hours during summer when you need relief most, but then loses its leaves in winter to let warming sun through when you want it—basically free air conditioning. Thermal mass (like thick concrete or masonry walls) is simple: it's material that slowly absorbs heat during the day and releases it at night, smoothing out temperature swings so your system doesn't work as hard. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING ****38.34586°N, 122.30466°W | Napa County, CA****

****Structural Design Drivers****

This site falls within IECC Zone 3C (Marine), with an Annual Freezing Index of 0°F-days, indicating negligible freeze-thaw cycling and no frost-driven loading. Wind, snow, and seismic parameters are not quantified in available data, preventing determination of which load case will govern the structural frame. The absence of these parameters is a critical data gap: in Napa County, seismic demand typically controls lateral design, but confirmation requires site-specific seismic hazard assessment and wind speed mapping per ASCE 7. Recommend immediate acquisition of USGS seismic hazard curves and recorded peak ground acceleration for the precise coordinates to establish whether lateral resistance or gravity load efficiency drives primary structural decisions.

****Geotechnical Foundation Requirements****

The site is underlain by Coombs gravelly loam on 2 to 5 percent slopes, classified as Hydro Group C with well-drained characteristics. Hydro Group C soils exhibit moderate infiltration capacity; the gravelly loam texture supports drainage but requires confirmation of the actual fines content fraction from a Phase I boring program. With zero frost depth, conventional frost-protected shallow foundations (FPSF) or standard shallow footings are viable without depth penalties. Drainage design must account for the moderate infiltration rate; subsurface water interception and positive site grading will be essential, particularly on slopes. No subsidence basin is recorded, removing settlement hazard from regional compaction, though bearing capacity and slope stability must be verified through geotechnical investigation before footing depth and width are finalized.

****Hazard Assessment****

No Quaternary faults are documented within the search radius, and no subsidence basin data exists for this location, eliminating those two hazards from preliminary risk screening. Seismic demand remains the primary structural unknown and must be quantified immediately. Wildfire risk data was not provided but should be cross-checked against Cal Fire hazard maps given Napa County's fire history; if the site falls within a high-hazard zone, defensible space, access, and water supply requirements will cascade into site planning and utility design.

****Site Context & Mining Proximity****

Three active mineral producers operate within a 5-mile radius: Basalt Rock Company (2 miles, clay), Mt George Pumice Deposit (3.1 miles, pumice), and Silva Pumice Quarry (5 miles, pumice). Blasting vibration, dust transport, and heavy vehicle traffic from these operations should be assessed for impact on building occupancy and amenity use. No active or producing oil and gas wells are present; the nearest well (Mobil Oil Corporation, 7.2 miles) is plugged and poses no operational risk. The site's marine climate envelope (AFI = 0°F-days) eliminates snow load and freeze-thaw as design drivers, simplifying enclosure and mechanical system selection.

****Bottom Line:**** Obtain USGS seismic hazard data and ASCE 7 design parameters for this coordinate pair, confirm soil bearing capacity and drainage through Phase I boring, and verify proximity to active mining operations and Cal Fire high-hazard fire zones before finalizing structural system and foundation strategy.



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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

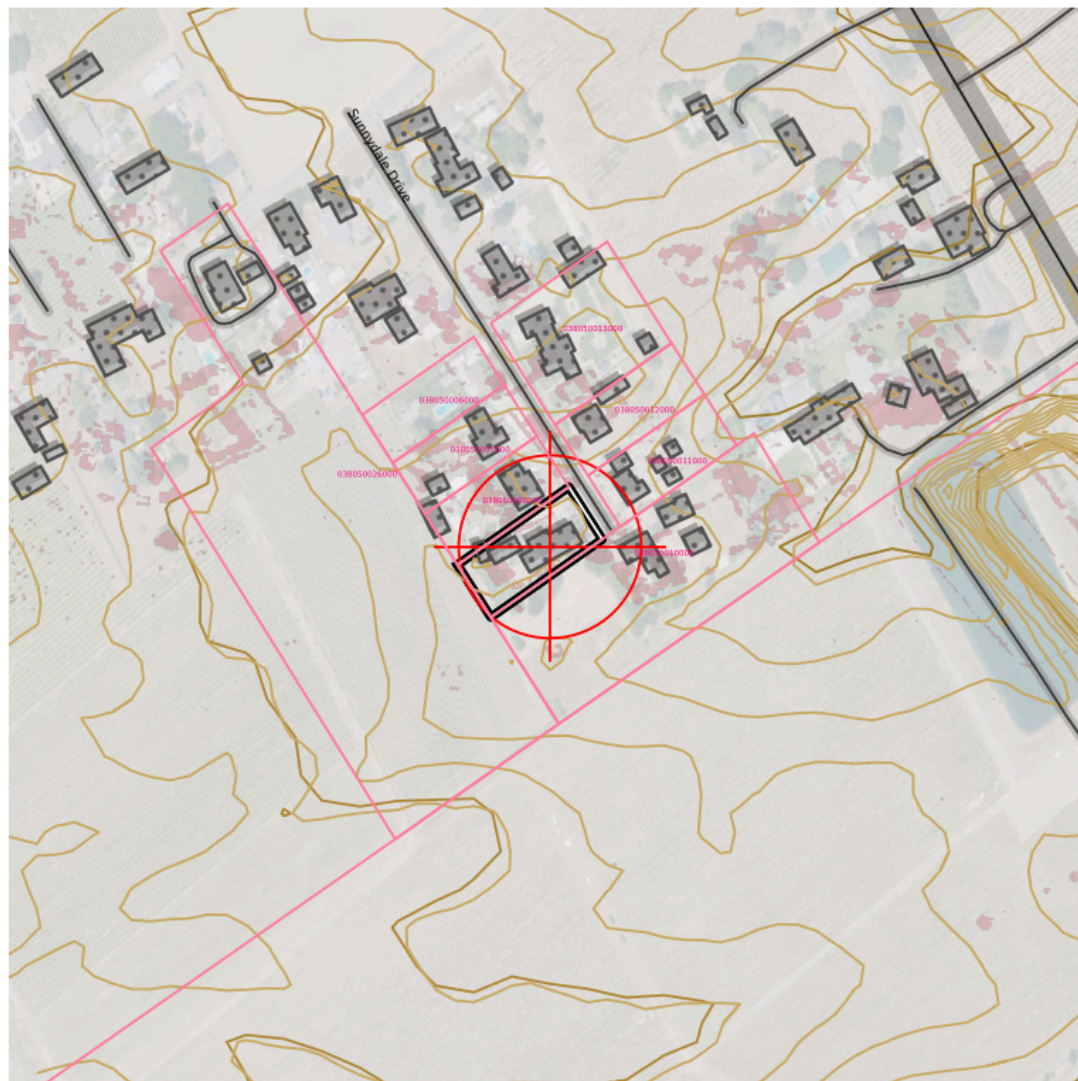


TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 12, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polyline (all geometry)

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – <https://geodesy.noaa.gov/CORS/cors-data.shtml>

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – <https://www.ngs.noaa.gov/cgi-bin/datasheet.prl>

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – [https://geodesy.noaa.gov/GEOID/](https://geodesy.noaa.gov/GEOID/GEOID18)

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

